



Semester Project Proposal

Sehat Saathi: A Hospital Management System for GIKI

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1 Introduction

This project aims to develop a comprehensive Hospital Management System (HMS) named “Sehat Saathi” for GIKI. The system will streamline hospital operations, improve patient care, and enhance administrative efficiency by automating key processes and providing a centralized platform for managing patient information, appointments, medical records, billing, and other essential functions. The modular architecture will allow for future expansion and integration with other healthcare systems.

1.1 Project Overview

Sehat Saathi will serve as an integrated platform connecting doctors, patients, administrators, and receptionists, facilitating seamless information flow and process automation within the hospital environment.

1.2 Objectives

- Develop a user-friendly and efficient HMS that meets the specific needs of a modern hospital
- Improve patient care by providing doctors with easy access to patient medical records
- Reduce operational costs by streamlining processes and minimizing paperwork
- Improve communication and collaboration among hospital staff
- Provide patients with convenient access to medical information and online appointment booking

2 Roadmap (Project Lifecycle)

2.1 Phase 1: Planning & Requirements Gathering

- Detailed analysis of hospital requirements
- Database schema design
- User interface design

2.2 Phase 2: Database Design

- Implementation of the database schema
- Data relationships and normalization

2.3 Phase 3: System Design & Development

- Implementation of Django models, views, and APIs
- Database integration
- Implementation of authentication and authorization
- Development of React components and user interfaces
- Integration with backend APIs

2.4 Phase 4: Testing & Debugging

- Thorough testing of all features and functionalities
- Unit testing for backend components
- Integration testing across the system

2.5 Phase 5: Deployment & Maintenance

- Deployment of the HMS to the production environment
- User training and documentation
- System monitoring and maintenance procedures
- Future enhancement planning

3 Functional Requirements

3.1 Doctor Module

- **Profile Access (via ID):** Manage personal profile, contact information, specialization, and department
- **Patient Medical Record Access:** Access complete patient medical records
- **Lab Test Ordering:** Order lab tests directly through the system

3.2 Patient Module

- **Appointment Booking:** Book appointments with preferred doctors
- **Available Doctor View:** View list of doctors and their specializations
- **Appointment Reminders:** Receive automated reminders about upcoming appointments
- **Medical Record Access:** Access personal medical records and download reports
- **Financial History View:** View financial history, including bills and payments

3.3 Administrator Module

- **User Management:** Perform CRUD operations on user accounts
- **Role-Based Permission Management:** Assign specific permissions to different roles
- **System Logs:** View system logs to track activities and errors
- **Statistics and Reporting:** Generate reports and view system statistics

3.4 Receptionist Module

- **Patient Registration:** Register new patients into the system
- **Appointment Management:** Create, read, update, and delete appointments
- **Bill and Invoice Generation:** Generate bills and invoices for patients
- **Available Doctor View:** View list of available doctors for scheduling
- **Patient Profile Updates:** Update patient profiles with new information

4 Non-Functional Requirements

4.1 Security & Compliance

- Data encryption for sensitive patient information
- Role-based access control mechanisms
- Compliance with healthcare data regulations
- Regular security audits and monitoring

4.2 Performance & Scalability

- Response time within 2 seconds for standard operations
- Support for concurrent users appropriate for hospital size
- Scalable architecture to accommodate growing patient data
- Efficient database queries and indexing

4.3 Usability & Accessibility

- Intuitive user interface for all user types
- Mobile-responsive design for access across devices
- Accessibility compliance with standard guidelines
- Comprehensive help documentation and tooltips

5 Project Management

5.1 Risk Management

- **Scope Creep:** Implement change management process to evaluate proposed changes
- **Technical Challenges:** Conduct thorough research and testing to address potential issues early
- **Integration Issues:** Plan for integration early and work closely with IT staff
- **Data Security:** Implement robust security measures including encryption and regular backups

5.2 Quality Assurance

- Regular code reviews and adherence to coding standards
- Comprehensive testing strategy (unit, integration, system testing)
- Continuous integration and delivery practices
- Feedback collection and iteration

6 Tools & Frameworks

6.1 Database Tools

- **PostgreSQL:** Primary relational database
- Database schema including tables for Patients, Doctors, Departments, Appointments, Medical Records, Prescriptions, Bills, DoctorAvailability, LabTests (This is subject to change)

6.2 Backend Frameworks

- **Django (Python web framework):** For server-side logic
- Django's built-in authentication and permission system

6.3 Frontend Frameworks

- **React:** JavaScript library for building user interfaces
- CSS frameworks for responsive design

6.4 Deployment & Monitoring

- To be determined based on specific requirements
- Options include cloud-based deployment (AWS, Azure, Google Cloud) or on-premise deployment
- Monitoring tools for system performance and availability

7 Conclusion

Sehat Saathi will provide a comprehensive solution to manage hospital operations efficiently, improving both patient care and administrative processes. By digitizing and automating key healthcare workflows, the system will reduce paperwork, minimize errors, and enhance communication between stakeholders. This Hospital Management System will serve as a foundational platform that can be expanded and integrated with other healthcare systems as GIKI's needs evolve. Upon successful implementation, we anticipate measurable improvements in operational efficiency, patient satisfaction, and resource utilization.